

DIABETIC PREDICTION FROM TONGUE IMAGE USING MACHINE LEARNING TECHNIQUES

MINI PROJECT REPORT

Submitted by

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22UEO013

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22UEO020

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22UEO027

YAMUNA.A

22UEO063

Under the guidance of

Mrs.GOKILAVANI

Head of the Department

In partial fulfillment of the requirement for the award of the Degree of

Bachelor of Engineering

in

Computer Science and Engineering

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF ENGINEERING**

**AVINASHILINGAM INSTITUTE FOR HOME SCIENCE AND
HIGHER EDUCATION FOR WOMEN**

COIMBATORE-641108

APRIL 2025

VISION AND MISSION



Avinashilingam Institute for Home Science and Higher Education for Women

(Deemed to be University under Category 'A' by MHRD, Estd. u/s3 of UGC Act 1956)

Re-accredited with 'A++' Grade by NAAC. Recognized by UGC under Section 12B, Coimbatore-
641043, Tamilnadu, India.

SCHOOL OF ENGINEERING

Approved by AICTE

Varapalayam, Thadagam(PO), Coimbatore-641108

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VISION

To empower women technocrats in the field of computer engineering and prepare the students for higher studies and professional career through technology, innovations and human values to serve the changing needs of our nation.

MISSION

- M1: To promote the all-round development of the students by providing them sound technical knowledge in the field of computer science and engineering based on the current advances in technology to help the development of nation.
- M2: To develop employability skills and research capability in women graduates through quality technical education.
- M3: To provide learning for the enhancement of technical skills, problem solving ability, team spirit, entrepreneurial skills with emphasis on value education with social ethics and cultural values in women technocrats.

CERTIFICATE

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This is to certify that the project report titled

‘Diabetic Prediction from Tongue Image Using Machine Learning techniques’

is being submitted by

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In partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering in Computer Science and Engineering, Department of Computer Science and Engineering , School of Engineering, Avinashilingam Institute for Home Science and Higher Education for Women, Coimbatore-641108.

This is the record of a bonafide work carried out by the above-mentioned students under our supervision and guidance. The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

Guide

Internal Examiner

External Examiner

(Mrs.Gokilavani)

Head of the Department

Dean

(Dr.S.Sivakumari)

(Dr.B.Sargunam)

PROGRAM OUTCOMES

Engineering Graduates will be able to:

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Conduct investigations of complex problems: Use research- based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multi disciplinary settings.

Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and

write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

The graduates in Computer Science and Engineering will be able to

PEO1: Design and develop novel products and innovative solutions for real life problems.

PEO2: Pursue higher studies and undertake Research & Development in Computer Science and Engineering.

PEO3: Excel as socially committed computer Engineers having effective communication skill, high ethical values and empathy for the needs of the society.

PROGRAMME SPECIFIC OUTCOMES (PSOs):

PSO1: Analyse and develop computer programs in the areas related Algorithms, database, web design, data mining, information Security cloud computing and networking for efficient design of Computer-based system of varying complexity.

PSO2: Gain knowledge in diverse areas of Computer Science and Experience an Environment conducive in cultivating skills for successful career, entrepreneurship and higher studies.

ACKNOWLEDGEMENT

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We express our heartfelt gratitude to our Registrar (i/c), **Dr.(Mrs.) H.Indhu Ph.D.,** Avinashilingam Institute for Home Science and Higher Education for Women, Coimbatore, for rendering her great support and being a footing stone of encouragement to us.

We extend our sincere thanks to our Dean, **Dr. (Mrs.) B.Sargunam M.E., Ph.D.,** School of Engineering. Avinashilingam Institute for Home Science and Higher Education for Women, Coimbatore, for her constant co operation and professional support in all our endeavors.

We feel very much grateful to **Dr. (Mrs.) S.Sivakumari, M.E., Ph.D.,** Professor, Head and our guide, Department of Computer Science and Engineering,

School of Engineering, for piloting us properly and stretching out a very big helping hand in the process of accomplishing our project.

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